Energy 258 (2022) 124913

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Thermochemical recuperation by steam methane reforming as an efficient alternative to steam injection in the gas turbines

Dmitry Pashchenko *, Ravil Mustafin, Igor Karpilov

Samara State Technical University, Molodogvardeiskaya 244 St, Samara 443100, Russia

ARTICLE INFO

Keywords:
Methane
Gas turbine
Thermochemical recuperation
Hydrogen-rich fuel
Thermodynamic analysis

ABSTRACT

Improving the energy efficiency of gas turbines is an extremely important task of the world energy industry. This paper presents the results of thermodynamic analysis of a gas turbine unit with thermochemical exhaust heat recuperation by steam methane reforming (chemically recuperated gas turbine — CRGT). The thermochemical exhaust heat recuperation systems consist of a reformer, a steam–methane mixture preheater, and a steam generator. The main goal of this paper is comparing CRGT efficiency with the efficiency of a gas turbine unit with steam injection (STIG). To calculate the recuperated heat in a reformer, the thermodynamic analysis of steam methane reforming was performed via Gibbs free energy minimization method. Both thermodynamic cycle and reforming analysis were performed via Aspen HYSYS. The analysis was performed in the temperature range of working fluid at a turbine inlet (T_{in}) of 800–1300 °C and the steam-to-methane ratio of 1, 2, 3 (mol-to-mol). The results showed that thermochemical exhaust heat recuperation can be considered as an alternative to steam injection in gas turbines. The efficiency of CRGT is higher than the efficiency of STIG for the same steam-to-methane ratio. With increasing in T_{in} the efficiency of CRGT is increasing. The efficiency of STIG and CRGT at T_{in}=1300 °C for a steam-to-methane ratio of 3 is 37.4% and 47.3%, respectively.

1. Introduction

Sustainable development of the world economy is inextricably linked with energy. At the same time, in the last century, fossil fuels have played a leading role in the energy sector. In the 21st century, despite advances in alternative and renewable energy sources, the share of fossil fuels in the global energy balance (according to World Energy Balance Highlights 2020 Edition presented by the International Energy Agency (IEA)) is about 64% [1]. According to the same highlights, the share of fossil fuels has decreased by 3% over the past ten years. However, while the production of oil and coal (with peat and oil shale) has changed insignificantly over ten years, natural gas production has increased from 2712 Mtoe to 3632 Mtoe. In addition, the IEA report "Golden Rules for a Golden Age of Gas" shows that natural gas (together with shale gas) will be the main source of energy in the 21st century for the global economy [2].

The problem of increasing the efficiency of electricity generation is the most important problem in the energy and power engineering industry. This problem has acquired extreme urgency in the face of constant growth in demand for electric energy. The widespread use of thermal power plants makes it necessary to constantly search for ways to increase their efficiency, because the increase in efficiency leads not only to fuel savings, but to a decrease in carbon dioxide emissions.

Gas turbine units (GTU) are one of the most common types of thermal power plants [3,4]. GTUs are widely used due to their high efficiency, relatively low specific capital investments, and ease of operation. The popularity of GTU in industry has led to a wide variety of its schemes [5]. Khan et al. performed a wide range of studies of various schematic diagrams of the gas turbines, e.g. regenerative gas turbine air-bottoming combined cycles [6–8], addition of new exchangers [9–12]. Kotowicz et al. proposed and analyzed various schemes of the combined cycle power plants with carbon dioxide capture [13,14]. Among the various ways to improve GTU efficiency, it is necessary to highlight the injection of steam (or water) into the combustion chamber, because this method is one of the simplest [15]. In this case, the increase in efficiency takes place mainly due to a decrease in the compressor output and an increase in the enthalpy of the high-temperature pressurized working fluid before a gas turbine [16].

There is a huge variety of different GTU schemes with steam (or water) injection (STIG) [17]. The first configuration of GTU with steam injection that has been realized in the industry is the scheme proposed by Armengaud and Lemale in 1906 [18,19]. The authors proposed using steam and water to reduce the gas temperature at the gas turbine inlet. GTUs with steam injection before the gas turbines have the abbreviation 'STIG'.

* Corresponding author.
E-mail address: pashchenkodmitry@mail.ru (D. Pashchenko).

https://doi.org/10.1016/j.energy.2022.124913
Received 10 January 2022; Received in revised form 24 June 2022; Accepted 21 July 2022
Available online 27 July 2022

Nomenclature

R	heat recuperation rate;
$r_{rec,i}$	share of the total recuperated heat in the i-th element of TCR;
H_i	enthalpy of i-th flow;
n_i	number of moles of i-th element;
β	steam-to-fuel ratio;
μ_i	chemical potential of element i;
G^t	value of total Gibbs free energy;
G^0_{fi}	standard Gibbs free energy of formation of element i;
R	molar gas constant;
T	temperature;
f_i	fugacity of pure elements i;
f^0_i	fugacity of pure elements i at the standard state;
C_{p_i}	specific heat of i component;
ΔH_{298}	standard enthalpy of the steam methane reforming reaction;
W_{GTU}	net work output of the gas turbine unit;
W_{gt}, W_{comp}	work of a gas turbine and compressor;
LHV	low heating value;
STIG	steam injection gas turbine;
CRGT	chemically recuperated gas turbine;
TCR	thermochemical recuperation.

Ahmed and Mohamed presented the results of the investigation of STIG where steam injection is taking place before the gas turbine inlet [20]. The authors showed that such a procedure leads to a significant increase (about 20%–30%) in the power output of GTU. Steam injection makes it possible to achieve the increase in energy efficiency even higher than in the combined cycles [21]. Lee et al. [22] investigated the effect of steam–water injection on the efficiency of microturbine for combined heat and power application. They performed an investigation via Aspen HYSYS. One of the key results is a significant increase in the efficiency of GTU by steam–water injection.

The schematic diagram of STIG is fairly simple. Steam (or water) is injected into either the combustion chamber or combustion products before the gas turbine. For the generation of steam, it is advisable to use the exhaust heat after the gas turbine. Therefore, the steam generation can be considered as an exhaust heat recuperation system [23]. The use of recuperation systems is also an important task to improve the efficiency of STIG.

It is widely known addition of steam in the combustion process leads to a decrease in NOx emission. Ivanov et al. suggested injecting steam into the methane stream for reduction of NOx emission in GTU [24]. According to the authors' statement, such way of steam injection also leads to notable improvement of the GTU efficiency.

If methane is premixed with steam in the STIG scheme, the initial mixture will be obtained for steam methane reforming — the main large-scale process for the production of synthesis gas and hydrogen [25,26]. Steam methane reforming is an endothermic process. Therefore, for its implementation, it is necessary to supply external heat. Exhaust heat can be used as such external heat. In this case, the recuperation of exhaust heat will be carried out not only due to the generation of steam but also due to the endothermic reaction of steam methane reforming. This method of exhaust heat recuperation is called thermochemical recuperation. In recent years, various studies on thermochemical recuperation for the different fuel-consuming equipment have been carried out: internal combustion engines, industrial furnaces, etc. Thermochemical recuperation can be considered as an alternative to steam injection into the combustion chamber.

A set of papers on GTU with thermochemical recuperation of exhaust heat were published during the last decades. GTU with thermochemical recuperation is called 'chemically recuperated gas turbines (CRGT)'. Olmsted and Grimes were the first who proposed using thermochemical recuperation based on steam methane reforming for the gas turbines [27]. The authors suggested partially replacing of steam superheater section with a steam methane reformer. The exhaust heat after a gas turbine was used for the endothermic steam methane reforming process. Then the products of reforming are fed into the turbine combustor. In that paper, Olmsted and Grimes showed that CH_4/H_2O absorbs exhaust heat thermally (as the temperature of mixture increases), and chemically (as the endothermic process takes place), resulting in a higher potential recuperation of exhaust heat than can be observed by conventional recuperation using air heating. Further, the idea of using TCR in gas turbines was found a wide discussion in the scientific literature.

Various schemes of TCR implementation in the GTU were proposed in last years [28–34]. Kesser et al. performed the thermodynamic analysis of a gas turbine power plant with thermochemical recuperation via steam methane reforming [35]. The efficiency of a gas turbine with TCR was compared with a steam-injected gas turbine, a combined cycle, and a simple gas turbine without any recuperation system. The authors determined that the gas turbines with TCR with a compressor pressure ratio of 30 have a thermal efficiency of about 47.0% while for a pressure ratio of 15, the thermal efficiency is about 47.3%. Both these variants are attractively high when for a steam-injected gas turbine, a combined cycle, and a simple gas turbine without any recuperation system.

Verkhivker and Kravchenko carried out the thermodynamic analysis of a combined cycle plant with thermochemical recuperation based on steam methane reforming [36]. The authors proposed using steam for reforming from a high-pressure cylinder of a steam turbine. Verkhivker and Kravchenko calculated the energy efficiency of a combined cycle plant with thermochemical recuperation has reached up to 90%. Additionally, the authors showed the positive impact of thermochemical recuperation on the environment. The use of synthesis gas after TCR leads to decreasing the amount of waste carbon dioxide ejected into the environment by nearly 20%.

Sadeghi et al. presented the results of thermoeconomic comparison and multi-objective optimization of combined gas turbine and organic Rankine cycle power generation system with thermochemical recuperation based on steam methane reforming and thermophysical recuperation [37]. The main goal of Sadeghi's paper is proposing and comparing (thermodynamically and economically) three power generation systems consisting of simple gas turbine/organic Rankine cycle, gas turbine/organic Rankine cycle with thermophysical recuperators, and gas turbine/organic Rankine cycle with thermophysical and thermochemical recuperators. The results showed that the thermodynamic, economic, and environmental efficiency of all systems is improved when the temperature of combustion products increases. The authors reported that the power generation system with thermophysical recuperators shows the best performance from the thermodynamic perspective with the 22.86% and 27.52% improvement in the exergy efficiency compared to the simple gas turbine/organic Rankine cycle and gas turbine/organic Rankine cycle with TCR, respectively. However, under the operating thermodynamic conditions obtained by the multi-objective optimization approach, the product cost associated with the system using TCR is less than that of the other configurations.

Moreover, the following fact has to be highlighted that it is not considered in the details in the present paper. In the gas turbines with thermochemical recuperation, hydrogen-rich fuel (synthesis gas) is used as a fuel for heat generation in a combustion chamber. There is sufficient information in the literature that replacing of original

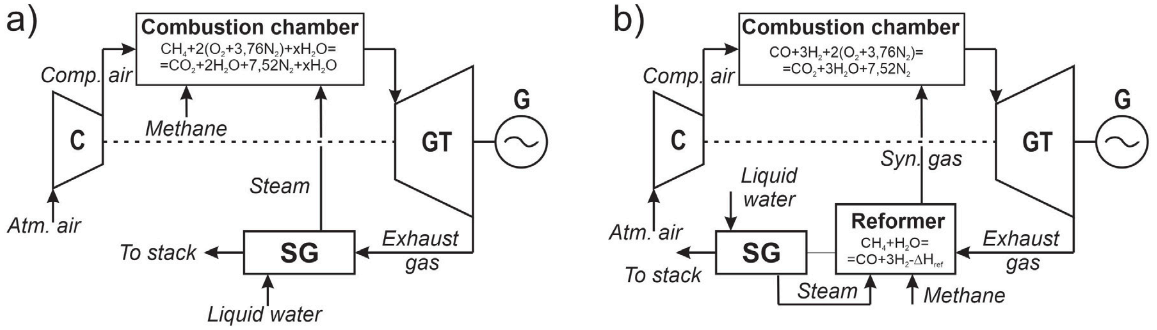

Fig. 1. The schematic diagram of heat generation from methane in STIG (a) and CRGT (b): C — compressor; GT — gas turbine; SG — steam generator; G — generator.

hydrocarbon fuel with the products of its reforming leads to decreasing in NOx emission [38–40]. Lui et al. presented the results of the investigation of methane reforming and combustion characteristics in thermochemically recuperated gas turbines [41]. The authors stated that TCR can be considered as ultralow NO_x emission technology.

The variety of GTU schemes and technical solutions makes the search for ways to improve efficiency an important task. The main goal of this work is to study a gas turbine with thermochemical recuperation by steam methane reforming from a point of view as an alternative solution to steam or water injection. To achieve this goal, a thermodynamic analysis of the gas turbine with thermochemical recuperation was performed and a comparison with steam and water injection was conducted. The efficiency of the thermochemical exhaust heat recuperation system strongly depends on the operational parameters of the steam–methane mixture (temperature, pressure, inlet steam-to-methane ratio); therefore, the thermodynamic analysis of GTUs was performed under various operating conditions. Moreover, the efficiency of the heat recuperation systems for both types of gas turbines was analyzed.

2. Methodology

2.1. Thermochemical recuperation concept

The main concept of thermochemical exhaust heat recuperation is using hot exhaust heat after a gas turbine for a thermochemical transformation of initial fuel, for example, natural gas via steam reforming [42,43]. Further, this transformed initial fuel in a new form (synthesis gas or syngas) is used in a combustion chamber. Since the reaction of natural gas steam reforming is strongly endothermic, external heat has to be supplied. In the thermochemical recuperation systems, the source of external heat is exhaust heat after the gas turbine.

The concept of thermochemical exhaust heat recuperation in GTU can be explained by comparison two types of GTU: STIG and CRGT. Fig. 1 shows the schematic diagram of heat generation from methane in STIG (a) and CRGT (b). As it can be seen from Fig. 1, the heat extraction from methane in STIG occurs in one stage — direct methane combustion in the combustion chamber. In CRGT the heat extraction from methane is divided into two stages. The first stage is a thermochemical transformation of initial methane by steam reforming reaction. In this stage, new synthetic fuel with combustion heat higher than combustion heat of initial methane (in comparison to 1 kg of initial methane) is obtained. And the second stage is the use of this new synthetic fuel for heat generation by combustion in a combustion chamber.

Another difference between STIG and CRGT is a more complete recuperation of exhaust heat. Exhaust heat after a gas turbine in STIG is used only for steam generation. However, in CRGT, exhaust heat is used not only for steam generation but also for steam methane reforming reaction. The schematic diagram in Fig. 1b can be slightly expanded by the addition of a preheater of the steam–methane mixture. Therefore, exhaust heat in CRGT is consecutively passed through a reformer, a preheater, and a steam generator. As a result, the heat recuperation rate in CTGT is higher than the heat recuperation rate in STIG.

One of the main criteria of the efficiency of the thermochemical recuperation systems is a heat recovery rate (HRR). HRR is a relation between the recuperated heat and the enthalpy of exhaust heat after a gas turbine. In STIG the recuperated heat is the heat transferred to H_2O in the steam generator. The recuperated heat for CRGT is a sum of heat that has been transferred from exhaust gas to steam–methane mixture in the reformer for the reforming process and the mixture preheater as well as heat for steam generation. In other words, the recuperated heat in CRGT can be determined as a sum of the enthalpy of the steam methane reforming process, change of the enthalpy of the steam–methane mixture by heating, and steam generation in the steam generator. Therefore, the heat recuperation rate can be determined via the following expression:

$$R = \frac{H_{rec}}{H_{exh}} = \frac{\Delta H_{298} + \Delta H_{mix} + \Delta H_{sg}}{H_{exh}} \tag{1}$$

where H_{rec} — a recuperated heat in the TCR system; H_{exh} — an exhaust gas enthalpy; ΔH_{298} — an enthalpy of the steam methane reforming process; ΔH_{mix} — the enthalpy change of the steam–methane mixture in a preheater; ΔH_{sg} — the enthalpy of steam generation.

Moreover, in this paper, the recuperated heat in a steam generator, a mixture preheater, a reformer as a share of the total recuperated heat in TCR system was determined from the following expression:

$$r_{rec,i} = \frac{H_{rec,i}}{H_{rec,tot}} \tag{2}$$

where $H_{rec,i}$ is the recuperated heat in the ith element of the scheme (steam generator for GTIG; reformer, mixture preheater, steam generator for CRGT); $H_{rec,total}$ is the total recuperated heat in whole TCR system.

The enthalpy of the steam methane reforming process in the expression (1) is depending on the operational parameters such as temperature, inlet steam-to-methane ratio, and pressure. Therefore, the thermodynamic analysis includes not only the cycle thermodynamics but also the thermodynamic of the steam methane reforming process.

2.2. Thermodynamic cycle analysis

The thermodynamic model has been developed and realized via Aspen HYSYS which is a powerful process simulation tool for schematic design of various concepts. Based on the developed model, parametric analysis was performed for each case of GTU - STIG and CRGT. The main goal of this analysis was to determine which operating parameters have a significant effect on overall GTU efficiency. There is a large number of publications on using Aspen HYSYS for modeling of various schemes of GTUs [44–48].

Thermodynamic analysis was performed for three schemes of GTU: steam injection with the recuperation of exhaust heat for steam generation (STIG); water injection without a recuperation; thermochemical

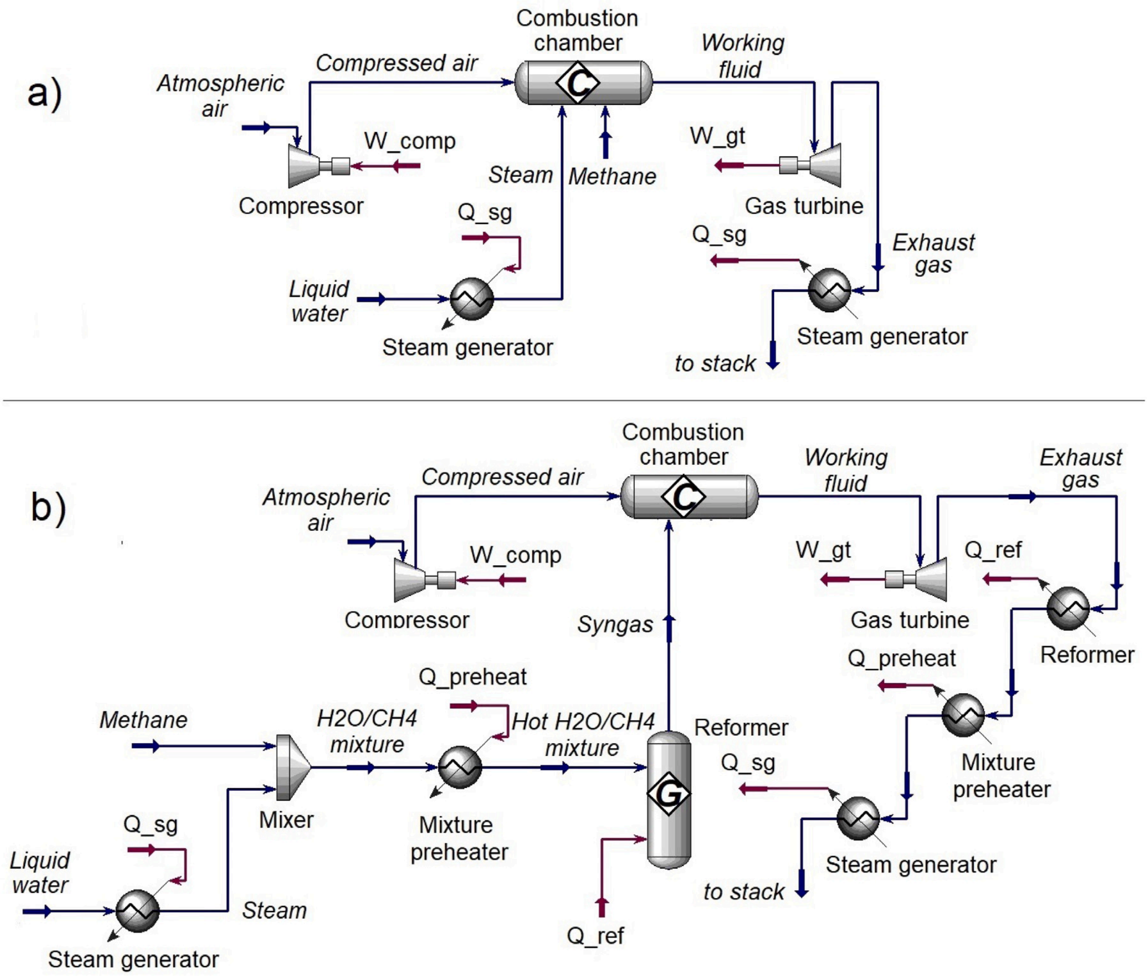

Fig. 2. The computational schemes of STIG (a) and CRGT (b) in Aspen HYSYS.

exhaust heat recuperation (CRGT). The calculation schemes in Aspen HYSYS are presented in Fig. 2. The schematic diagram of GTU with water injection is not shown because it is almost similar to STIG.

In this paper, the net work output of the gas turbine has been calculated based on the following expression:

$$W_{GTU} = W_{gt} - W_{comp} \tag{3}$$

where W_{gt} is the power produced in gas turbine; W_{comp} is the work for air compressing (compressor work).

In order to make a good comparative analysis, the working fluid temperature at the gas turbine inlet has been set as constant for all analyzed schemes. The temperature range of working fluid at the gas turbine inlet is from 800 to 1300 °C. The mass flow rate of methane is 100 kg/h for all analyzed schemes. The temperature of working fluid at the gas turbine inlet has been regulated by the air mass flow rate. Aspen HYSYS has an useful function, (*Adjust*), in which the user sets the target variable value and specifies an adjusted variable used to reach the target value. The temperature of exhaust gas has been set as the target variable and the air mass flow as the adjusted variable.

2.3. Thermodynamic analysis of SMR

The enthalpy of the steam methane reforming (SMR) process in the expression (1) is depending on the operational parameters such as temperature, inlet steam-to-methane ratio, and pressure. In addition, these operational parameters affect the composition of synthesis gas after a reformer. In the thermodynamic analysis of SMR, the equilibrium conditions are analyzed. Therefore, the type of catalysts, the type of reformer, etc is not playing a role in the reforming products and the enthalpy of the process. To determine the SMR process enthalpy and synthesis gas composition as a function of the operational parameters, the thermodynamic analysis via Gibbs free energy minimization method was performed.

The thermodynamic analysis makes it possible the calculation of the equilibrium concentration of the reforming products at the given operating parameters such as temperature, inlet steam-to-methane mixture and pressure. The main concept of the Gibbs free energy minimization method is the fact the chemical system is thermodynamically favorable when its total Gibbs free energy is minimum and its differential is equal to zero [49,50]:

$$dG^t = 0 \tag{4}$$

where G^t is total Gibbs free energy.

Total Gibbs free energy of the SMR decomposition process can be calculated as a sum of the chemical potential of all components:

$$G^t = \sum n_i \mu_i \tag{5}$$

where n_i is number of ith component moles; μ_i is chemical potential of ith component.

A chemical potential of each component can be calculated based on following expression:

$$\mu_i = \Delta G^0_{fi} + RT \ln\left(\frac{f_i}{f_i^0}\right) \tag{6}$$

where G^0_{fi} is standard Gibbs free energy of ith component formation; R is molar gas constant; T is system temperature; f_i is fugacity of a pure

Table 1
The comparison of the synthesis gas composition at a steam-to-methane ratio of 2, % (mol/mol).

Temperature	Pressure	Source	Gas composition, %				
			H_2	CO	H_2O	CO_2	CH_4
1000 K	1 bar	HYSYS	63.55	14.22	16,11	5.22	0.9
		Verkhivker [36]	63.42	14.43	16.11	5.12	0.92
		Classical [52]	63.01	14.32	16.32	5.23	1.12
1000 K	5 bar	HYSYS	61.08	14.85	17.57	5.13	1.55
		Verkhivker [36]	61.44	14.93	17.97	4.18	1.48
		Classical [52]	60.66	14.82	17.32	5.55	1.65
1100 K	1 bar	HYSYS	63.78	15.95	16,2	3.98	0.09
		Verkhivker [36]	63.76	16.11	16.11	3.94	0.08
		Classical [52]	63.88	16.08	16.02	3.93	0.09
1100 K	5 bar	HYSYS	61.24	14.71	18,11	4.28	1.66
		Verkhivker [36]	61.45	14.93	17.96	4.18	1.48
		Classical [52]	61.28	15.02	17.87	4.24	1.59

ith component; f_i^0 is fugacity of a pure ith component at the standard state.

If the equations presented above will be combined, the equation for total Gibbs free energy of the chemical system can be written as the following:

$$G^t = \sum n_i \Delta G^0_{fi} = \sum n_i RT \ln\left(\frac{f_i}{f_i^0}\right) \tag{7}$$

The thermodynamic analysis of SMR was performed also via Aspen HYSYS software that is a widely used program in chemical engineering for the modeling of chemical systems similar to the considered thermochemical recuperation system. To simulate the steam methane reforming process, the ideal Gibbs reactor is chosen. The Gibbs reactor determines the reforming products by the Gibbs free energy minimization method of each component based on the chemical mechanism of the SMR process. The Peng–Robinson equation of state was used to calculate the volume of the product gas as a function of pressure and temperature [51].

3. Results and discussion

3.1. Verification

Although Aspen HYSYS is a generally recognized tool for modeling chemical processes, verification of the results of thermodynamic analysis of the SMR process was carried out in this article. The obtained results were compared with the results of Verkhivker and Kravchenko [36]. Moreover, the results were compared with the results of a classical thermodynamic analysis based on a conjugate solution of material balance and law action mass equations [52].

The thermodynamic analysis of the SMR process was carried out for various operating parameters. A comparison of the results of the synfuel composition is shown in Table 1. The upper results are calculated via Aspen HYSYS, the middle rows are the mole fraction from the paper presented by Verkhivker and Kravchenko [36], the bottom results are the mole fraction of each component that was calculated based on the classical thermodynamic analysis [52]. A good correlation is observed for all results.

Moreover, the comparison of the thermodynamic analysis results with the results of other authors [53,54] showed a good correlation for the wide range of the operational conditions.

3.2. Cycle analysis

The initial conditions and turbine parameters for a parametric analysis are presented in Table 2.

It should be noted that in the real gas turbines the gas turbine expander from the temperature of at turbine inlet 1000 °C and higher must be cooled (e.g. open air - through the air bleed from the compressor) [55,56]. A steam mass flow rate was variable in the models of STIC and CRGT. A working fluid temperature was controlled by the air mass flow rate. For example, an increase in steam mass flow rate leads to a decreasing in the mass flow rate of air.

Table 2
Stream properties and efficiency parameters of gas turbine and compressor.

Parameter	Value
CH_4 mass flow rate	100 kg/h
Air mass flow rate	1600–2200 kg/h
Gas temperature at turbine inlet	800–1300 °C
Turbine isentropic efficiency	90%
Compressor isentropic efficiency	85%
Reforming temperature	T_{exh}-20 °C
Steam temperature	185.4 °C
Steam mass flow rate	112.5; 225; 337.5 kg/h
Inlet temperature of water	20 °C
Temperature of atmospheric air	20 °C
Pressure of all inlet streams (except air)	10 bar

The temperature of steam is the temperature of saturated steam at a pressure of 10 bar. The mass flow rate of steam is 112.5; 225; 337.5 kg/h. These values of steam mass flow rates are equal to the steam-to-methane ratios (mol-to-mol) of 1; 2; 3 at the reformer inlet for the CRGT scheme. Therefore, the same mass flow rates of steam are chosen for the STIG scheme.

Fig. 3 show the net work output as a function of working fluid temperature at the turbine inlet for four analyzed schemes of the gas turbine units: GTU without heat recuperation and injection (black line); GTU with water injection (green lines); STIG (brown lines); CRGT (red lines). Fig. 3 presents the results for various mass flow rates of steam (or water): 112.5 kg/h (solid points); 225 kg/h (asterisk); 337.5 kg/h (hollow points).

The GTU without any recuperation and steam injection systems was chosen as the reference value of the net work output. Fig. 3a shows that the addition of liquid water to the combustion chamber leads to a significant decrease in the net power. This is due to the fact that a significant amount of heat in the combustion chamber is spent on the evaporation of water. Also, after the gas turbine, an exhaust gas heat recuperation system is not installed. Therefore, although water injection reduces the compressor work leads to a decrease in the net work output.

The steam injection to the combustion chamber (STIG) leads to an increase in the net work output. This is due to the fact that the exhaust heat is used to generate steam. Moreover, the steam injection leads to a decrease in air mass flow rate and, as a result, to a decrease in the compressor work. The increase in the steam mass flow rate leads to a proportional increase in the net work output. Fig. 3 shows that steam injection is an effective way to increase the efficiency of the GTUs.

However, the maximum increase in the net work output takes place in the GTU with the thermochemical exhaust heat recuperation system (CRGT). The net work output in CRGT at a temperature of working fluid as the turbine inlet of 1300 °C is 25% more than the net work output of STIG. Fig. 3a shows that the net work output of CRGT increases for the entire temperature range from 800 to 1300 °C . At the same time, the net work output of STIG increases insignificantly in the temperature range above 1100 °C. This is due to the fact that with an increase in the inlet temperature, the temperature of the exhaust gases increases. And further, with an increase in the exhaust gas temperature, the enthalpy of the steam methane reforming process increases. Therefore, the recuperation system is recovering more exhaust heat to the GTU cycle. Fig. 3b shows that in the temperature range up to 1000 °C, the increase in the net work output is insignificant and is mainly caused by the heating of methane as a component of the steam–methane mixture. However, with an increase in the exhaust gas temperature, the net work output of the CRGT increases due to an increase in the enthalpy of

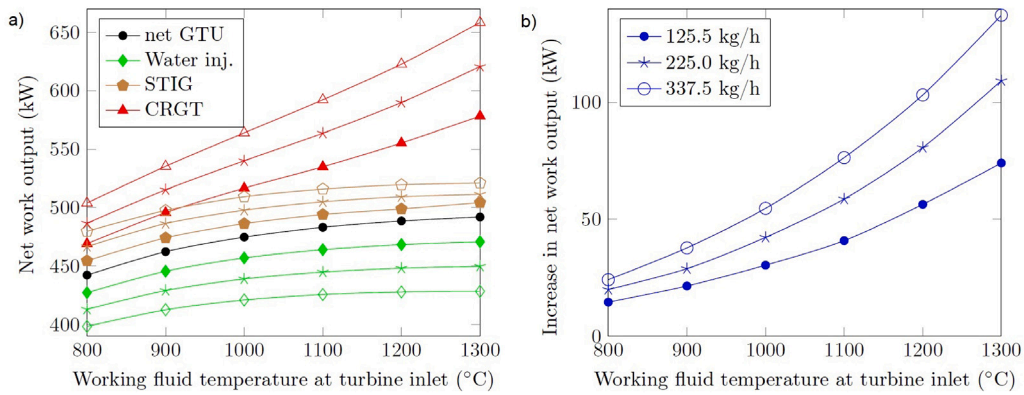

Fig. 3. (a) The net work output for GTU without heat recuperation and without injection (black line); GTU with water injection (green lines); STIG (brown lines); CRGT (red lines). Mass flow rate of water at GTU inlet 112.5 kg/h (solid points); 225 kg/h (asterisk); 337.5 kg/h (hollow points); b) Increase in the net work output from STIG to CRGT.

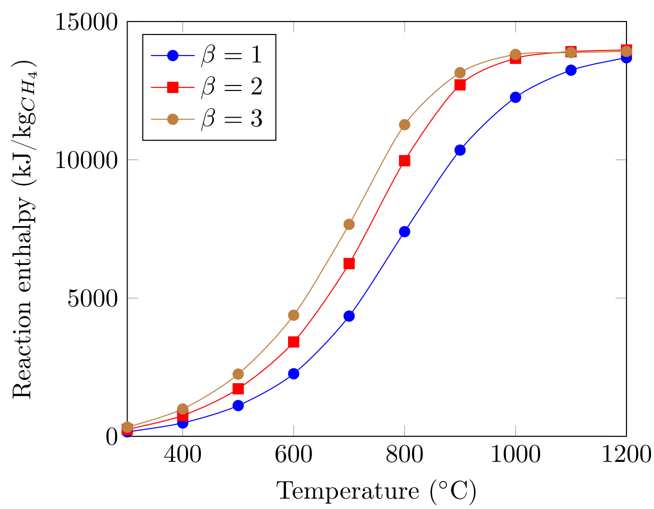

Fig. 4. The enthalpy of the steam methane reforming process as a function of a temperature at the steam-to-methane ratios (β) of 1, 2, 3 at a pressure of 10 bar.

the steam methane reforming process. In addition, for the steam-to-methane ratio of 3, the increase in the net work output between STIG and CRGT is greater than for the steam-to-methane ratio of 1. This is due to the fact that an increase in the steam-to-methane ratio leads to an increase in the methane conversion and, as a consequence, to an increase in the enthalpy of the process.

To understand the effect of temperature and steam-to-methane ratio in the net work output, the enthalpy of the steam methane reforming process was calculated via Aspen HYSYS. Fig. 4 shows the enthalpy of the steam methane reforming process as a function of temperature at the steam-to-methane ratios (β) of 1, 2, 3 at a pressure of 10 bar.

Fig. 4 shows that in the temperature range before 600 °C the process enthalpy is slightly increasing. In the temperature range of 600–800 °C the process enthalpy is drastically increasing. And in the temperature range above 800 °C is reaching the maximum value of 13970 kJ/kg. Fig. 4 makes it clear the effect of exhaust gas temperature and steam-to-methane ratio on the net work output of CRGT.

Fig. 5 shows the recovered heat in the elements of the thermochemical exhaust heat recuperation system: in a steam generator, in a mixture preheater, in a reformer for the analyzed CRGT.

The recuperated heat in the steam generator is constant according to the analysis conditions (Table 2). The recuperated heat in the mixture preheater monotonically increases with an increase in the exhaust gas temperature. This is due to the fact that, according to the analysis conditions, the steam–methane mixture temperature depends on the exhaust gas temperature. The recuperated heat in the reformer at a temperature of 800 °C is minimal. However, at a temperature of 1300 °C, there is more heat used to generate steam (for a steam-to-methane ratio of 1). This is due to the fact that at a high temperature of the exhaust gas, the enthalpy of the process increases. For a ratio of 3, the recuperated heat in the steam generator is maximum for all analyzed temperatures.

The recuperated heat in a steam generator, a mixture preheater, a reformer as a share of the total recuperated heat in the thermochemical recuperation system at the steam-to-methane ratios of 1, 2, 3 is presented in Fig. 6.

Fig. 6 shows that the share of the recuperated heat in a reformer is increasing with temperature increasing. At the same time, the share of the recuperated heat in a steam generator is decreasing when temperature increases. This is due to the fact that the recuperated heat in a steam generator is constant when the recuperated heat in a mixture preheater and a reformer is increasing.

One of the main advantages of thermochemical exhaust heat recuperation is a transformation of exhaust heat into the chemical energy of synthetic fuel (syngas). In other words, if exhaust heat is supplied to the reaction mixture, its temperature is not increasing because all supplied heat is converted to chemical energy. To understand the effect of thermochemical transformation of exhaust heat, the temperature of exhaust gases was compared. Fig. 7 shows the temperature of exhaust gas after a gas turbine and the temperature of exhaust gas to stack as a function of working fluid temperature at a gas turbine inlet. Temperatures were compared for CRGT (a) and STIG (b).

An exhaust gas temperature is almost the same for all analyzed cases, the discrepancy between the temperatures is up to 10 °C for various cases. Therefore, the temperature of the exhaust gases is shown in the single red lines. Fig. 7a depicts an effect of thermochemical transformation of exhaust heat into chemical energy. In the temperature range before 900 °C the enthalpy of the steam methane reforming process is minimal and the increase in temperature of exhaust after a gas turbine leads to a directly proportional increase in temperature of exhaust gas to stack. However, if an exhaust gas temperature is increasing above 900–1000 °C, an increase in the temperature of exhaust gas is slowing down because a part of exhaust heat is transformed into chemical energy.

In addition, Fig. 7 shows that an exhaust gas has a potential for s steam generation for a steam turbine cycle. Therefore, CRGT can be combined with a steam turbine to obtain a chemically recuperated combined cycle plant.

To show that thermochemical recuperation can be considered as an alternative to steam injection in the gas turbines, the efficiency of STIG and CRGT was compared. The efficiency of a gas turbine unit [57] was calculated based on the following expression:

$$\eta_{GTU} = \frac{W_{gt} - W_{comp}}{m_{CH_4} \cdot LHV_{CH_4}} \tag{8}$$

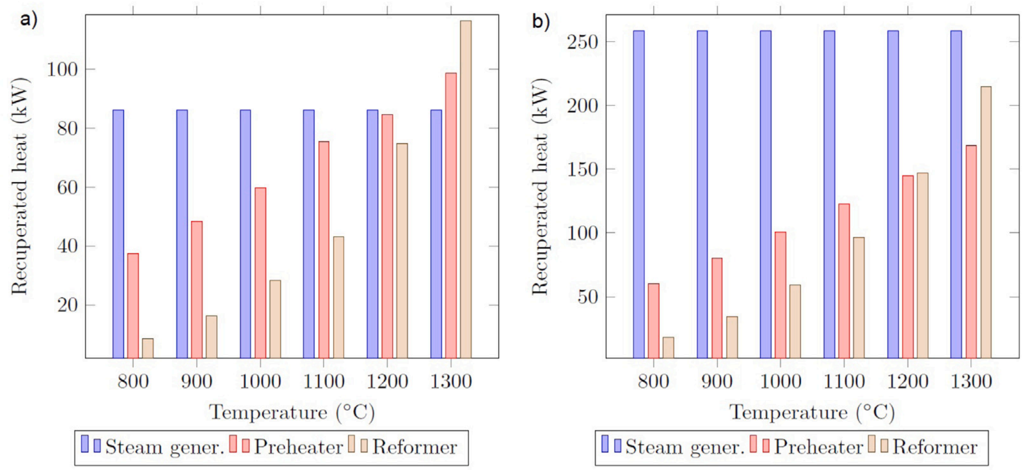

Fig. 5. The recuperated heat in the thermochemical recuperation system for the steam-to-methane ratio of 1 (a) and 3 (b) at various temperatures of a working fluid at a turbine inlet.

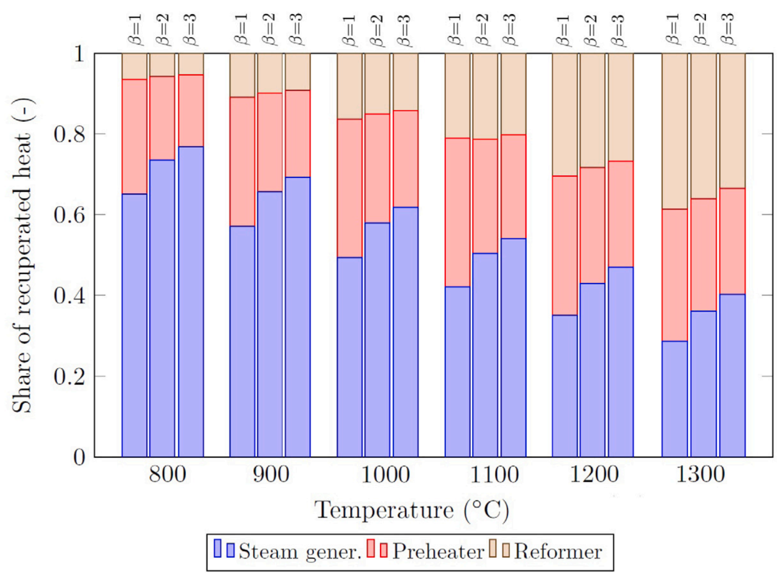

Fig. 6. The recuperated heat in a steam generator, a mixture preheater, a reformer as a share of the total recuperated heat in the TCR system at the steam-to-methane ratios of 1, 2, 3.

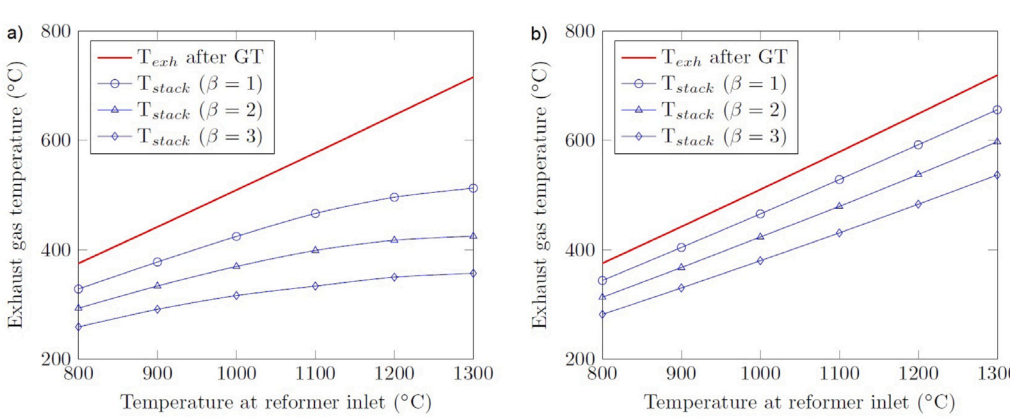

Fig. 7. A temperature of exhaust gas after a gas turbine and temperature of exhaust gas to stack as a function of working fluid temperature at a gas turbine inlet: a) CRGT; b) STIG.

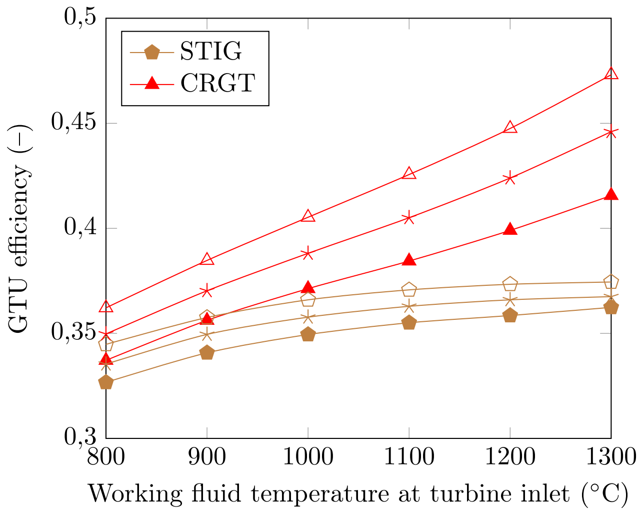

Fig. 8. The efficiency of STIG (a) and CRGT (b) as a function of temperature of working fluid at a gas turbine inlet for various steam mass flow rates: 112.5 kg/h (solid points); 225 kg/h (asterisk); 337.5 kg/h (hollow points).

where m_{CH_4} is the mass flow rate of methane; LHV_{CH_4} is the methane lower heating value.

Fig. 8 shows that CRGT is more efficient in comparison with STIG. The maximum efficiency has for the high-temperature gas turbines. The efficiency of STIG and CRGT at T_{in}=1300 °C for a steam-to-methane ratio of 3 is 37.4% and 47.3%, respectively. The main reason for an increase in the efficiency of CRGT at T_{in}=800 °C is preheating of methane as a part of a steam–methane mixture. With an increase in T_{in} the recuperated heat in a reformer is increasing. Therefore, a gas turbine with thermochemical exhaust heat recuperation is more efficient than STIG.

Based on Fig. 8 we can conclude that thermochemical exhaust heat recuperation by steam methane reforming can be considered as an efficient alternative to steam injection in the gas turbines.

4. Conclusion

The comparative thermodynamic analysis of various schemes of the gas turbine units (water injection, steam injection (STIG), thermochemical recuperation (CRGT)) was performed. The thermodynamic cycle analyses were done under various operating parameters: inlet temperature of 800–1300 °C; a steam-to-methane ratio of 1, 2, 3 (mol-to-mol); operating pressure of 10 bar. The thermodynamic analysis showed that CRGT has the highest efficiency in comparison with other schemes (STIG, water injection). The net work output in CRGT at the temperature of working fluid as the turbine inlet of 1300 °C is 25% more than the net work output of STIG. To understand the effect of thermochemical transformation of exhaust heat in a reformer, the enthalpy of the steam methane reforming process was calculated via Gibbs free energy minimization method. In the temperature range before 600 °C the process enthalpy is slightly increasing. In the temperature range of 600–800 °C the process enthalpy is drastically increasing. And in the temperature range above 800 °C is reaching the maximum value of 13970 kJ/kg. The recovered heat in the elements of the thermochemical exhaust heat recuperation system was determined as well as the recuperated heat in a steam generator, a mixture preheater, a reformer as a share of the total recuperated heat in the TCR system at the steam-to-methane ratios of 1, 2, 3. CRGT is more efficient in comparison with STIG. The maximum efficiency has for the high-temperature gas turbines. The efficiency of STIG and CRGT at T_{in}=1300 °C for a steam-to-methane ratio of 3 is 37.4% and 47.3%, respectively. Therefore, thermochemical exhaust heat recuperation by steam methane reforming can be considered as an efficient alternative to steam injection in the gas turbines.

CRediT authorship contribution statement

Dmitry Pashchenko: Methodology, Writing, Conceptualization. **Ravil Mustafin:** Modeling, Visualization. **Igor Karpilov:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

This work is supported by the Russian Science Foundation under grant 19-19-00327.

References

[1] International Energy Agency. World energy balances: Overview. International Energy Agency Paris; 2020.
[2] International Energy Agency. Golden rules for a golden age of gas: world energy outlook special report on unconventional gas. OECD/IEA; 2012.
[3] Boyce Meherwan P. Gas turbine engineering handbook. Elsevier; 2011.
[4] Poullikkas Andreas. An overview of current and future sustainable gas turbine technologies. Renew Sustain Energy Rev 2005;9(5):409–43.
[5] Jansohn P. Overview of gas turbine types and applications. In: Modern gas turbine systems. Elsevier; 2013, p. 21–43.
[6] Khan MN, Alkanhal Tawfeeq Abdullah, Majdoubi Jihen, Tlili Iskander. Performance enhancement of regenerative gas turbine: air bottoming combined cycle using bypass valve and heat exchanger—energy and exergy analysis. J Therm Anal Calorim 2021;144(3):821–34.
[7] Khan MN. Energy and exergy analyses of regenerative gas turbine air-bottoming combined cycle: optimum performance. Arab J Sci Eng 2020;45(7):5895–905.
[8] Khan Mohammad Nadeem. Energetic and exergetic investigation of regenerative gas turbine air-bottoming/steam-bottoming combined cycle. Mechanics 2021;27(3):251–8.
[9] Khan MN, Tlili I, Khan WA. Thermodynamic optimization of new combined gas/steam power cycles with HRSG and heat exchanger. Arab J Sci Eng 2017;42(11):4547–58.
[10] Khan MN, Tlili I. Performance enhancement of a combined cycle using heat exchanger bypass control: A thermodynamic investigation. J Cleaner Prod 2018;192:443–52.
[11] Khan MN, Tlili I. New approach for enhancing the performance of gas turbine cycle: A comparative study. Results Eng 2019;2:100008.
[12] Khan MN, Tlili I. New advancement of high performance for a combined cycle power plant: thermodynamic analysis. Case Stud Therm Eng 2018;12:166–75.
[13] Kotowicz Janusz, Brzkeczek Mateusz, Job Marcin. The influence of carbon capture and compression unit on the characteristics of ultramodern combined cycle power plant. Int J Global Warming 2017;12(2):164–87.
[14] Kotowicz Janusz, Brzeczek Mateusz Andrzej. The influence of CO_2 capture and compression on the economic characteristics of a combined cycle power plant. J Power Technol 2013;93(5):314.
[15] Larson Eric D, Williams Robert H. Steam-injected gas turbines. 1987.
[16] Carapellucci Roberto, Milazzo Adriano. Repowering combined cycle power plants by a modified STIG configuration. Energy Convers Manage 2007;48(5):1590–600.
[17] Najjar Yousef SH. Gas turbine cogeneration systems: a review of some novel cycles. Appl Therm Eng 2000;20(2):179–97.
[18] Meyer Adolf. The combustion gas turbine: its history, development, and prospects. Proc Inst Mech Eng 1939;141(1):197–222.
[19] Eckardt Dietrich, Rufli Peter. Advanced gas turbine technology: ABB/BBC historical firsts. In: Turbo expo: power for land, sea, and air. vol. 78521, American Society of Mechanical Engineers; 2001, V003T02A010.
[20] Ahmed Mahmoud Salem, Mohamed Hany Ahmed. Performance characteristics of modified gas turbine cycles with steam injection after combustion exit. Int J Energy Res 2012;36(15):1346–57.
[21] Bhargava Rakesh K, Bianchi Michele, Campanari Stefano, De Pascale Andrea, Negri di Montenegro Giorgio, Peretto Antonio. A parametric thermodynamic evaluation of high performance gas turbine based power cycles. J Eng Gas Turbines Power 2010;132(2).
[22] Lee Jong Jun, Jeon Mu Sung, Kim Tong Seop. The influence of water and steam injection on the performance of a recuperated cycle microturbine for combined heat and power application. Appl Energy 2010;87(4):1307–16.

[23] Nishida Kousuke, Takagi Toshimi, Kinoshita Shinichi. Regenerative steam-injection gas-turbine systems. Appl Energy 2005;81(3):231–46.
[24] Ivanov AA, Ermakov AN, Shlyakhov RA. On high suppression of NO x and CO emissions in gas-turbine plants with combined gas-and-steam cycles. Therm Eng 2010;57(13):1132–8.
[25] Yan Yongliang, Thanganadar Dhinesh, Clough Peter T, Mukherjee Sanjay, Patchigolla Kumar, Manovic Vasilije, et al. Process simulations of blue hydrogen production by upgraded sorption enhanced steam methane reforming (SE-SMR) processes. Energy Convers Manage 2020;222:113144.
[26] De Falco Marcello, Santoro Giulia, Capocelli Mauro, Caputo Giampaolo, Giaconia Alberto. Hydrogen production by solar steam methane reforming with molten salts as energy carriers: Experimental and modelling analysis. Int J Hydrogen Energy 2021;46(18):10682–96.
[27] Olmsted JH, Grimes PG. Heat engine efficiency enhancement through chemical recovery of waste heat. In: 7th Intersociety energy conversion engineering conference. 1972.
[28] Janes Jack. Chemically recuperated gas turbine. California Energy Commission, Energy Technology Development Division ...; 1992.
[29] Bolland O, Stadaas JF. Comparative evaluation of combined cycles and gas turbine systems with water injection, steam injection, and recuperation. 1995.
[30] Briesch MS, Bannister RL, Diakunchak IS, Huber DJ. A combined cycle designed to achieve greater than 60 percent efficiency. 1995.
[31] Newby Richard A, Yang Wen-Ching, Bannister Ronald L. Use of thermochemical recuperation in combustion turbine power systems. In: Turbo expo: power for land, sea, and air. vol. 78699, American Society of Mechanical Engineers; 1997, V002T05A008.
[32] Botros KK, de Boer MJ, Fletcher HG. Thermodynamics, environmental and economic assessment of CRGT for exhaust heat recovery in remote compressor station applications. In: ASME 1997 international gas turbine and aeroengine congress and exhibition. American Society of Mechanical Engineers Digital Collection; 1997.
[33] Carcasci Carlo, Facchini Bruno, Harvey Simon. Modular approach to analysis of chemically recuperated gas turbine cycles. Energy Convers Manage 1998;39(16–18):1693–703.
[34] Pashchenko Dmitry, Mustafin Ravil, Karpilov Igor. Ammonia-fired chemically recuperated gas turbine: Thermodynamic analysis of cycle and recuperation system. Energy 2022;252:124081.
[35] Kesser Keith Franklin, Hoffman MA, Baughn JW. Analysis of a basic chemically recuperated gas turbine power plant. J Eng Gas Turbines and Power 1994;116(2):277–84.
[36] Verkhivker Gregoriy, Kravchenko Vladimir. The use of chemical recuperation of heat in a power plant. Energy 2004;29(3):379–88.
[37] Sadeghi Mohsen, Chitsaz Ata, Marivani Parisa, Yari Mortaza, Mahmoudi SMS. Effects of thermophysical and thermochemical recuperation on the performance of combined gas turbine and organic rankine cycle power generation system: Thermoeconomic comparison and multi-objective optimization. Energy 2020;210:118551.
[38] Poran Arnon, Tartakovsky Leonid. Energy efficiency of a direct-injection internal combustion engine with high-pressure methanol steam reforming. Energy 2015;88:506–14.
[39] Poran A, Tartakovsky L. Performance and emissions of a direct injection internal combustion engine devised for joint operation with a high-pressure thermochemical recuperation system. Energy 2017;124:214–26.
[40] Tartakovsky Leonid, Baibikov Vladimir, Veinblat Mark. Comparative performance analysis of SI engine fed by ethanol and methanol reforming products. Technical report, SAE Technical Paper; 2013.
[41] Liu Xiao, Zheng Hongtao, Liu Qian. Research of methane reforming and combustion characteristics in chemically recuperated gas turbine. Ind Eng Chem Res 2014;53(5):1940–6.
[42] Popov SK, Svistunov IN, Garyaev AB, Serikov EA, Temyrkanova EK. The use of thermochemical recuperation in an industrial plant. Energy 2017;127:44–51.
[43] Pashchenko Dmitry. Natural gas reforming in thermochemical waste-heat recuperation systems: A review. Energy 2022;123854.
[44] Liu Zuming, Karimi Iftekhar A. Simulating combined cycle gas turbine power plants in Aspen HYSYS. Energy Convers Manage 2018;171:1213–25.
[45] Pashchenko Dmitry. Industrial furnaces with thermochemical waste-heat recuperation by coal gasification. Energy 2021;221:119864.
[46] Pashchenko Dmitry. Thermochemical waste-heat recuperation as on-board hydrogen production technology. Int J Hydrogen Energy 2021;46(57):28961–8.
[47] Canepa Roberto, Wang Meihong. Techno-economic analysis of a CO_2 capture plant integrated with a commercial scale combined cycle gas turbine (CCGT) power plant. Appl Therm Eng 2015;74:10–9.
[48] Pashchenko Dmitry, Mustafin Ravil, Karpilov Igor. Efficiency of chemically recuperated gas turbine fired with methane: Effect of operating parameters. Appl Therm Eng 2022;212:118578.
[49] Aktaş Seda, Karakaya Mustafa, Avcı Ahmet K. Thermodynamic analysis of steam assisted conversions of bio-oil components to synthesis gas. Int J Hydrogen Energy 2009;34(4):1752–9.
[50] Özkara-Aydınoğlu Şeyma. Thermodynamic equilibrium analysis of combined carbon dioxide reforming with steam reforming of methane to synthesis gas. Int J Hydrogen Energy 2010;35(23):12821–8.
[51] Peng Ding-Yu, Robinson Donald B. A new two-constant equation of state. Ind Eng Chem Fundam 1976;15(1):59–64.
[52] Pashchenko Dmitry. Thermodynamic equilibrium analysis of steam methane reforming based on a conjugate solution of material balance and law action mass equations with the detailed energy balance. Int J Energy Res 2020;44(1):438–47.
[53] Tabrizi Farshad Farshchi, Mousavi Seyed Amir Hossein Seyed, Atashi Hossein. Thermodynamic analysis of steam reforming of methane with statistical approaches. Energy Convers Manage 2015;103:1065–77.
[54] Lutz Andrew E, Bradshaw Robert W, Keller Jay O, Witmer Dennis E. Thermodynamic analysis of hydrogen production by steam reforming. Int J Hydrogen Energy 2003;28(2):159–67.
[55] Mishra Shivam, Sharma Abhishek, Kumari Anupam, et al. Response surface methodology based optimization of air-film blade cooled gas turbine cycle for thermal performance prediction. Appl Therm Eng 2020;164:114425.
[56] Sahu Mithilesh Kumar, et al. Investigation of the effect of air film blade cooling on thermoeconomics of gas turbine based power plant cycle. Energy 2016;115:1320–30.
[57] Kotowicz Janusz, Job Marcin, Brzkeczek Mateusz, Nawrat Krzysztof, Mkedrych Janusz. The methodology of the gas turbine efficiency calculation. Arch Thermodyn 2016;(4).